

Maendeleo ya Ubepari nchini Indonesia

Zhihu • 2026-08-08

Nakumbuka mwaka jana wakati maandamano makubwa yalipotokea Indonesia, Nepal na nchi nyingine, mtu alichapisha makala akijiuliza "wapi njia ya watu wa nchi za nusu-koloni nusu-feudal"^[1]. Nchi ya nusu-feudal kwa ujumla ni nchi ya kilimo ambayo haijakamilisha viwanda, lakini Indonesia baada ya miongo kadhaa ya maendeleo inaonekana haiko tena nchi ya kilimo. Zaidi ya hayo, kutokana na maoni yangu ya kusafiri Indonesia, kiwango cha maendeleo ya nchi hii kinakaribia kiwango cha majimbo ya kati ya China katika miaka ya 2000; ni vigumu kuihusisha na picha ya jamii ya nusu-koloni nusu-feudal inayopambana nje ya mlango wa viwanda. Makala hii inachunguza maendeleo ya ubepari wa Indonesia katika ngazi mbalimbali.

I. Kiwango cha maendeleo ya uchumi wa Indonesia

1. Viashiria kuu vya uzalishaji wa viwanda

Kama inavyoonyeshwa kwenye picha hapa chini, isipokuwa chuma na saruji, ambazo ziko nyuma na zinalingana na kiwango cha China mwanzoni mwa miaka ya 1990, uzalishaji wa bidhaa kuu za viwanda vya Indonesia kwa sasa umefikia au karibu kufikia kiwango cha China mwanzoni mwa karne ya 21.

国家(年份)	总人口 (百万)	百万人钢产量 (吨)	百万人煤炭产 量(吨)	人均发电量(千 瓦时)	百万人炼油 量(吨)	百万人汽车产 量	百万人PVC产 量(吨)	百万人水泥 产量(吨)
印尼2025	288	65973	283681	1584	177778	3985	1737	224653
中国1990	1134	58509	95239	551	94357	449	693	184930
中国1995	1218	78293	106569	770	115764	1191	1129	366175
中国2000	1265	100582	79052	993	166799	1623	1354	450937
中国2005	1304	272846	167945	1782	219326	4372	4979	812884
中国2010	1340	467646	241792	2944	315672	13627	8433	1394030

Kwa sasa, Indonesia inajaza mapungufu ya viwanda vizito kwa kuvutia mtaji wa kigeni.

Mwaka 2025, uwezo wa Indonesia wa chuma cha pua ulizidi tani milioni 5, ikawa nchi ya pili duniani; kwa kila mtu imezidi kiwango cha China 2025. Makampuni kuu yanayozalisha chuma cha pua nchini ni makampuni ya China na makampuni ya Asia Kusini. Makampuni ya Korea pia yanapanga kujenga kiwanda cha chuma cha pua cha tani milioni 2 nchini Indonesia pamoja na makampuni ya China^[2].

Katika usafishaji wa mafuta, miradi inayoendelea ni pamoja na viwanda viwili vya kiserikali au vya pamoja^{[3][4][5]}.

Katika umeme, katika miaka 10 ijayo Indonesia itaongeza karibu gigawati 70 za uwezo wa uzalishaji, zaidi ya nusu ikiwa umeme wa kijani^[6]; uzalishaji wa kila mtu unaweza kuongezeka takriban kilowati-saa 900, na kufikia kilowati-saa 2000 kwa mwaka — kiwango cha China 2008.

Kwa ufupi, kutoka data ya uzalishaji wa viwanda, Indonesia tayari imefikia kiwango cha nchi ya viwanda, lakini viwanda bado havijakamilika; nchi iko katika mchakato wa viwanda wa haraka, na inatarajiwa kuhitaji takriban miaka 15 kufikia kiwango cha China 2012 na kukamilisha viwanda kwa kiasi.

2. Kiwango cha miundombinu

Indonesia ni nchi ya visiwa; wakazi wamesambaa kwenye makumi ya maelfu ya visiwa, na hali za kujenga miundombinu mikubwa ni mbaya zaidi kuliko China. Lakini wakazi wamejikita hasa katika visiwa vya Java na Sumatra, vikichukua karibu 60% na zaidi ya 20% ya wakazi wa kudumu, pamoja zaidi ya 80% ya nchi. Eneo la Java karibu 140,000 km², wakazi milioni 140, kiwango cha juu cha uchumi; linaweza kulinganishwa na mkoa wa Guangdong (eneo 180,000 km², wakazi milioni 100). Sumatra ni kubwa (470,000 km²) lakini uchumi wake iko nyuma; inaweza kulinganishwa na mkoa wa Sichuan (480,000 km²).

(1) Kulinganisha Java na Guangdong

Mwaka 2025, urefu wa jumla wa barabara za Java kilomita 120,000 — kiwango cha Guangdong 2005 (kilomita 115,337). Barabara za haraka za Java kilomita 1782 — kiwango cha Guangdong 2002 (kilomita 1885).

Mwaka 2025, reli za Java karibu kilomita 4700 — kiwango cha Guangdong 2015 (kilomita 4021). Reli ya kwanza ya mwendo kasi ya Java — Jakarta-Bandung — tayari inafanya kazi, kilomita 142. Ya kwanza ya Guangdong — Wuhan-Guangzhou — ilianza mwishoni mwa 2009.

Mwaka 2025, Java ina mstari mmoja wa metro wa kilomita 15.7; mistari miwili ya reli nyepesi (pamoja na ile inayounganisha miji ya satelaiti ya Jakarta) jumla ya kilomita 50. Mwaka 2004, Guangdong ilikuwa na kilomita 59 za metro, hakuna reli nyepesi.

Mwaka 2025, bandari kubwa ya Java — Tanjung Priok ya Jakarta — itafikia TEU milioni 8 — kiwango cha bandari ya Shenzhen 2002 (TEU milioni 7.61).

Kwa ujumla, miundombinu ya Java imefikia kiwango cha mkoa wa juu zaidi wa China katikati ya miaka ya 2000.

(2) Kulinganisha Sumatra na Sichuan

Mwaka 2025, barabara za Sumatra karibu kilomita 93,000 — kiwango cha Sichuan 2000 (kilomita 91,000). Barabara pekee ya kiwango cha juu ni barabara ya kulipia ya Trans-Sumatra (JTTS); kufikia mwisho wa 2025 kilomita 1235 zimefunguliwa — kiwango cha Sichuan 2003 (kilomita 1501).

Mwaka 2025, reli za Sumatra kilomita 1340 tu — kiwango cha Sichuan 1965 (kilomita 1530). Sumatra haina reli za mwendo kasi; Sichuan ilifungua ya kwanza 2010.

Mwaka 2025, Sumatra nzima ina mstari mmoja wa reli nyepesi ya mjini (Palembang), kilomita 23.4. Sichuan ilifungua mstari wa kwanza wa metro 2010, kilomita 18.5. Kwa hiyo, kwa usafiri wa reli, Sumatra inakaribia kiwango cha Sichuan 2010.

Kwa ujumla, miundombinu ya Sumatra imefikia kiwango cha majimbo ya magharibi ya China miaka ya 2000, lakini ujenzi wa reli uko nyuma sana.

Kwa Java na Sumatra, zinazochukua 80% ya wakazi wa kudumu, kiwango cha miundombinu kinakaribia kiwango cha China miaka ya 2000. Kwa sasa, serikali ya Indonesia inaendeleza kwa nguvu ujenzi wa barabara, reli na bandari; miradi mikuu inayoendelea:

- Barabara ya Trans-Sumatra: urefu wa mipango kilomita 2818 (sehemu kadhaa za awamu ya kwanza na ya pili zimefunguliwa; sasa ujenzi unaendelea kwa kilomita 1000 zilizobaki), inapangwa kukamilika 2031.
- Reli kuu ya Trans-Sumatra: ujenzi wa zaidi ya kilomita 1000 za nyimbo mpya kuunganisha reli za zamani zilizotawanyika (kilomita 1200). Awamu ya kwanza: Banda Aceh-Besitang, kilomita 478, kabla ya 2030.
- Mradi wa kuongeza reli ya Jakarta-Bandung (Jakarta-Bandung-Yogyakarta-Surabaya): karibu kilomita 570; kwa sababu ya uwekezaji mkubwa, inatarajiwa kukamilika miaka ya 2030.
- Mstari wa metro ya Jakarta mashariki-magharibi (awamu ya tatu): kilomita 84.1 (Banten, Jakarta, Java Magharibi), inapangwa kufunguliwa 2031.
- Awamu ya pili ya metro ya Jakarta kaskazini-kusini: kilomita 11.8 (2A: 5.8, 2B: 6.0), inapangwa 2029.
- Bandari ya kina ya Patimban (awamu ya pili na tatu): TEU milioni 7.5 (Indonesia na Japan), inapangwa 2027.
- Bandari mpya ya Tanjung Priok (upanuuzi wa Kalibaru): TEU milioni 15-18, inapangwa 2030.

3. Viashiria vya fedha

Mwaka 2025, uzalishaji wa jumla wa viwanda vya Indonesia dola bilioni 276.5; kwa uwiano wa 0.61 hadi dola 2005, ni dola bilioni 168.665 za 2005. Mwaka 2005, uzalishaji wa viwanda vya China dola bilioni 736. Wakazi wa Indonesia 2025 milioni 288; wakazi wa China 2005 bilioni 1.3. Kwa hiyo, uzalishaji wa viwanda vya Indonesia kwa uwiano wa wakazi ni sawa na dola bilioni 761.3 za kiwango cha China 2005.

Uuzaji wa nje wa bidhaa za Indonesia 2025 dola bilioni 282.91; kwa uwiano wa 0.61 ni dola bilioni 172.5 za 2005. Uuzaji wa China 2005 dola bilioni 762. Kwa uwiano wa wakazi, uuzaji wa Indonesia 2025 ni sawa na dola bilioni 778.6 za kiwango cha China 2005.

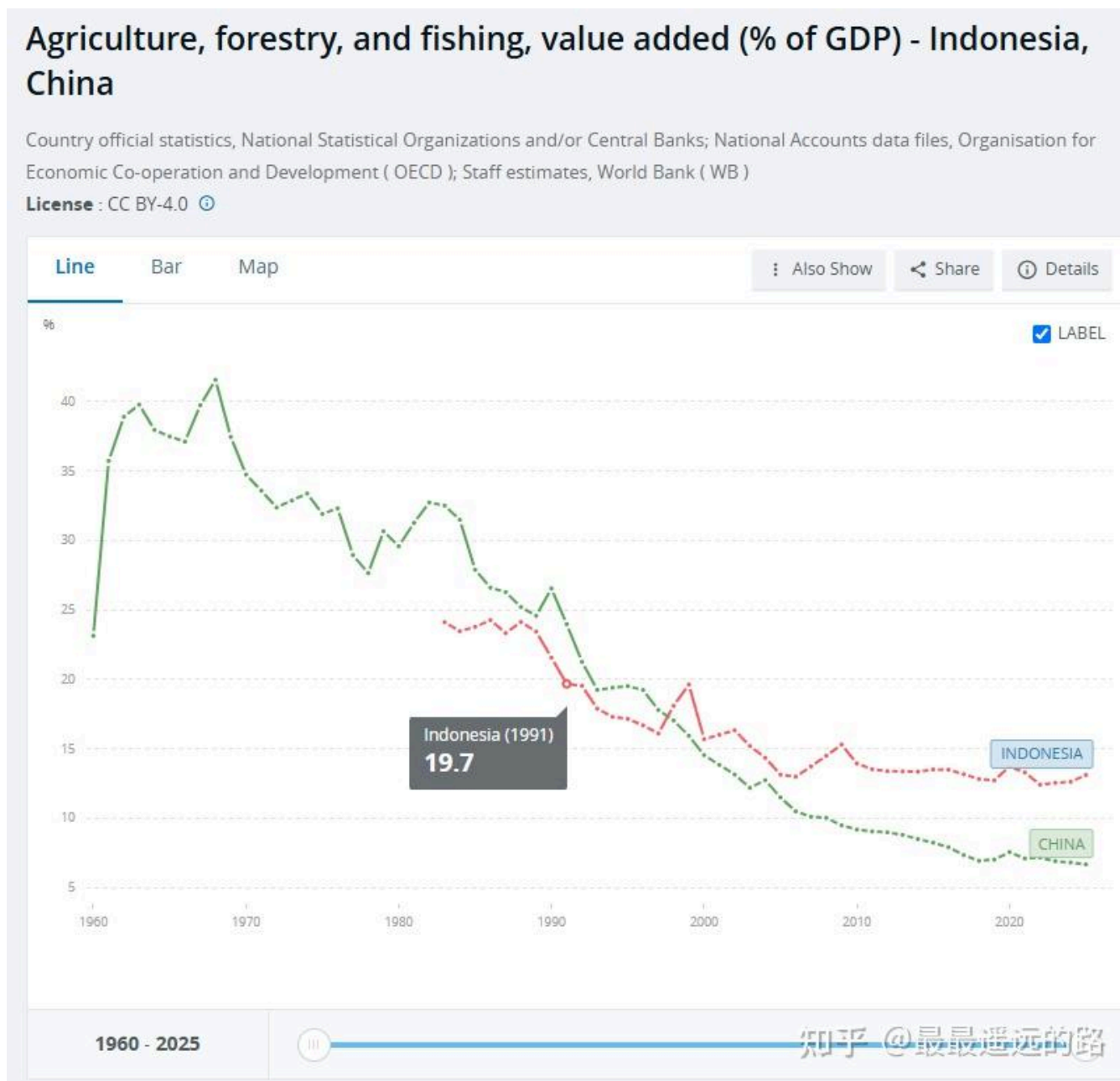
Vivyo hivyo, GDP ya kila mtu ya Indonesia 2025 dola 5083; kwa uwiano wa 0.67 hadi dola 2008, ni dola 3405. GDP ya kila mtu ya China 2008 karibu dola 3300. Kwa hiyo, GDP ya kila mtu ya Indonesia 2025 inakaribia kiwango cha China 2008.

Kwa ujumla, viashiria vya fedha vya Indonesia vinalingana na viashiria vya uzalishaji na miundombinu, na vinakaribia kiwango cha China katikati ya miaka ya 2000.

II. Muundo wa viwanda na tabaka la Indonesia

1. Je, Indonesia ni nchi ya viwanda?

Mwaka 1990, sehemu ya kilimo katika GDP ya Indonesia ilishuka chini ya 20%^[7].



Agriculture, forestry, and fishing, value added (% of GDP) - Indonesia, China

Kwa viwanda, mwanzoni mwa miaka ya 2000 sehemu yake katika GDP ililingana na China, zaidi ya 30%^[8]. Baada ya China kuingia WTO, viwanda vya Asia ya Kusini-Mashariki vilipigwa, na sehemu ya viwanda katika GDP ilipungua. Lakini katika miaka ya hivi karibuni, kwa uhamishaji mkubwa wa viwanda vya China nje, sehemu hii imetulia na kuongezeka kidogo. Lakini tangu miaka ya 1990, sehemu ya viwanda katika GDP ya Indonesia imekuwa juu kuliko kilimo; pamoja na ujenzi, viwanda vyote viko juu zaidi ya kilimo.

Manufacturing, value added (% of GDP) - Indonesia, China

Country official statistics, National Statistical Offices (NSOs); National Accounts data files, Central Banks; Staff estimates, World Bank (WB)

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Manufacturing, value added (% of GDP) - Indonesia, China

Kwa hiyo, kwa kigezo cha nchi ya viwanda kilichotumika China — uzalishaji wa viwanda unazidi wa kilimo — Indonesia ni nchi ya viwanda tangu miaka ya 1990. Kwa "kigezo cha Chenery", uzalishaji wa kilimo wa Indonesia umeshuka chini ya 20%, karibu 10%, ikiwa katika nusu ya pili ya kiwango cha kati cha viwanda; karibu na kukamilika (chini ya 10%).

2. Uwiano wa proletaria: Indonesia 2025 dhidi ya Urusi ya Tsarist 1913

Katika Urusi ya Tsarist 1913^{[9][10]}, uwiano wa watu wa umri wa kufanya kazi karibu 52%, jumla milioni 88. Proletaria yote isiyo ya kilimo milioni 15 — 17% ya watu wa umri wa kufanya kazi. Kwa viwanda na ujenzi tu, karibu milioni 7 — 8%. Pamoja na wafanyakazi milioni 6.5 wa kilimo^[11], jumla ya wafanyakazi milioni 21.5 — 24.4% ya wafanyakazi wote.

Mining	645,000
Construction	1,600,000
Small industries	3,706,000
Commerce	865,000
Household	3,000,000
Menial laborers	2,500,000
State-owned armaments, shipbuilding, etc.	1,500,000
Approximate total	15,000,000

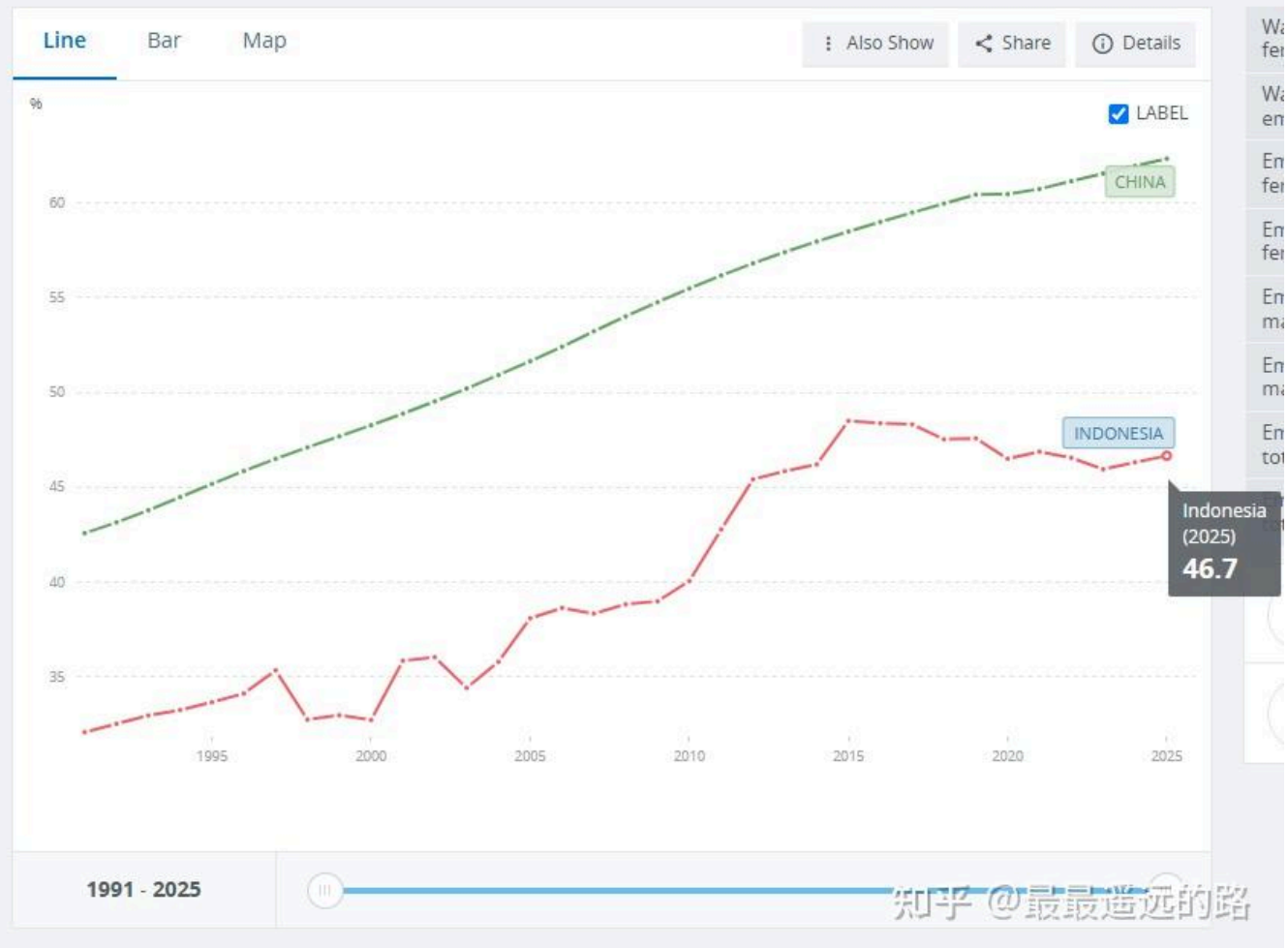
Table 1: Hired Non-Agricultural Manual Labor, Usually Omitted Categories (early 1914)

Uwiano wa wafanyakazi wa kukodiwa katika Indonesia unakaribia 50%^[12].

Wage and salaried workers, total (% of total employment) (modeled ILO estimate) - Indonesia, China

ILO Modelled Estimates database (ILOEST), International Labour Organization (ILO), uri: ilostat.ilo.org/data/bulk, publisher: ILOSTAT, type: external database, date accessed: January 17, 2026

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Wage and salaried workers, total (% of total employment) (modeled ILO estimate) - Indonesia, China

Wafanyakazi wa viwanda wa Indonesia sasa ni karibu 23% ya wafanyakazi wote — kiwango cha China 1995–2005^[13].

Employment in industry (% of total employment) (modeled ILO estimate) - Indonesia, China

ILO Modelled Estimates database (ILOEST), International Labour Organization (ILO), uri: ilostat.ilo.org/data/bulk, publisher: ILOSTAT, type: external database, date accessed: January 17, 2026

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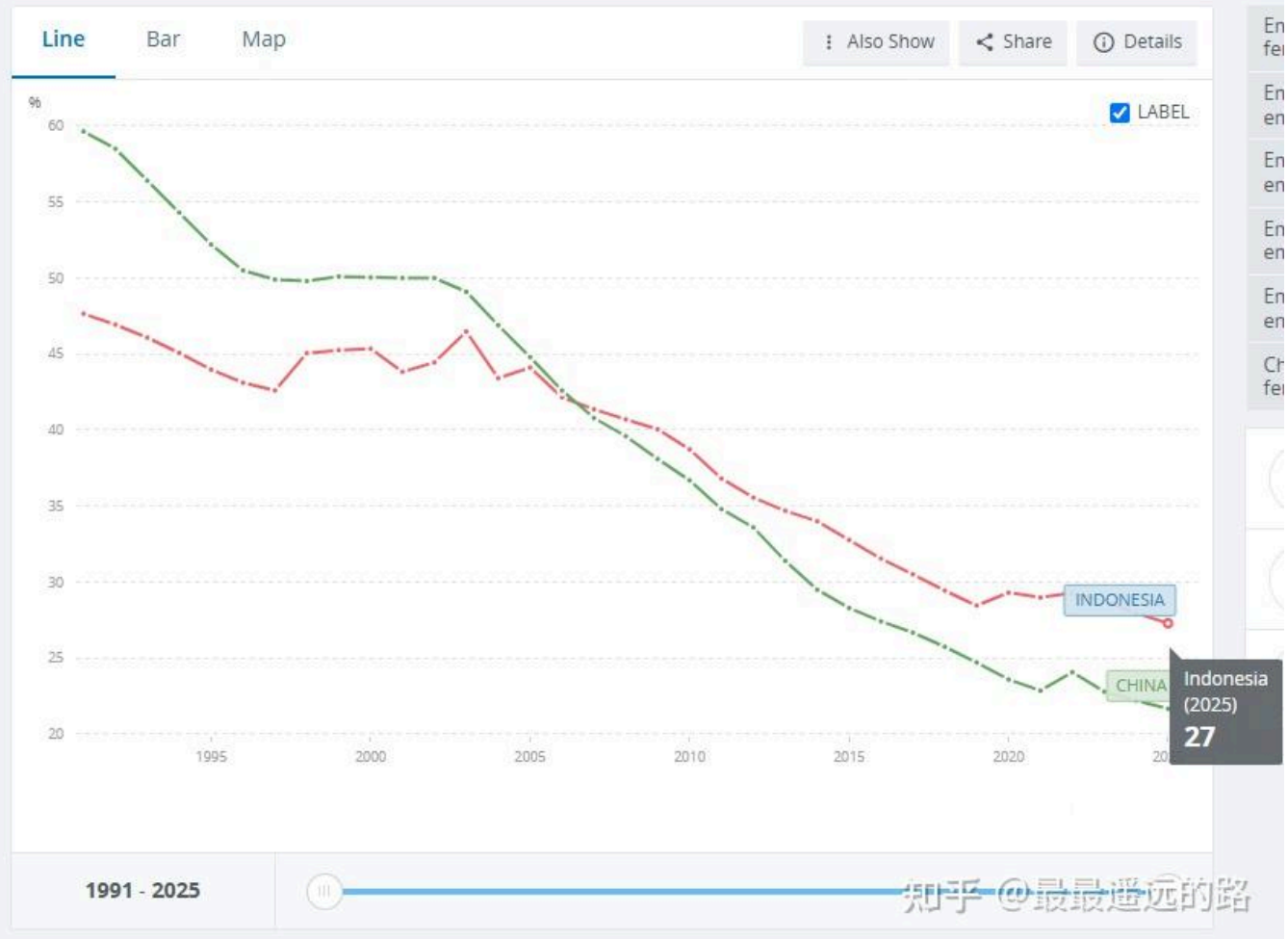
Employment in industry (% of total employment) (modeled ILO estimate) - Indonesia, China

Wafanyakazi wa kilimo wa Indonesia sasa ni 27% tu^[14].

Employment in agriculture (% of total employment) (modeled ILO estimate) - Indonesia, China

ILO Modelled Estimates database (ILOEST), International Labour Organization (ILO), uri: ilostat.ilo.org/data/bulk, publisher: ILOSTAT, type: external database, date accessed: January 17, 2026

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Employment in agriculture (% of total employment) (modeled ILO estimate) - Indonesia, China

Kwa hiyo, Indonesia 2025 — kwa uwiano wa proletaria, proletaria ya viwanda, na wafanyakazi wa kilimo — ni ya kibepari zaidi kuliko Urusi ya Tsarist 1913, na kwa ujumla karibu na Ufaransa na Ujerumani 1900. Kwa muundo halisi wa tabaka, kuiona Indonesia kama nchi ambayo haijakamilisha ubepari ni kinyume sana na ukweli.

3. Mabaki mangapi ya feudal yanabaki katika kilimo cha Indonesia?

Kulingana na data ya tafiti za wasomi wa Indonesia, wakulima wengi wa Indonesia ni wamiliki wa ardhi zao^[15].

Table 5.2 Land management and sharecropping.

Region	Own	Lease	Sharecropping	Ratio
Sumatra	14,452 (23.39%)	5,826 (36.90%)	3,101 (33.16%)	2
Java	24,708 (39.99%)	4,976 (31.52%)	3,319 (35.49%)	3
Kalimantan	6,646 (10.76%)	1,243 (7.87%)	829 (8.87%)	4
Sulawesi	7,535 (12.20%)	1,414 (8.96%)	676 (7.23%)	4
Bali & Nusa Tenggara	7,064 (11.43%)	2,222 (14.07%)	1,270 (13.58%)	3
Maluku	747 (1.21%)	72 (0.46%)	57 (0.61%)	6
Papua	632 (1.02%)	35 (0.22%)	99 (10.6%)	5
Indonesia	61,784 (70.75%)	45,788 (18.08%)	9,351 (10.71%)	3

Source: Statistics Indonesia (BPS), 2014.

知乎 @最最遥远的路

Kwa kulinganisha, Japan 1941 ilikuwa na 30% ya wakulima wasio na ardhi kabisa, na 40% wengine walihitaji kukodi sehemu ya ardhi^[16].

第2表 農地改革前後における自小作別農家戸数の変化 (都府県), 1941-50年					
年	農家戸数(千戸)				
	合計 ^a	自作 ^b	自小作	小自作	小作
1941	5,203 (100)	1,589 (31)	1,095 (21)	1,080 (21)	1,439 (28)
1947	5,700 (100)	2,068 (36)	1,157 (20)	985 (17)	1,489 (26)
1950	5,895 (100)	3,652 (62)	1,569 (27)	403 (7)	271 (5)

出典：農林省(1952, pp. 55-60).
注：^a例外規定農家(土地を耕作しない)を除く。
^b耕作地主を含む。

知乎 @最最遥远的路

Kwa muundo wa umri, vijana ni wachache kuliko wazee; katika muongo uliopita, mwelekeo ni vijana kuondoka kilimo kwenda miji kufanya kazi^[17].

Table 3. Number of individual agricultural holdings by age group in Indonesia, ST 2013 and ST 2023 (household)

Age group (years)	ST 2013		ST 2023		Growth (%)
	HH	%	HH	%	
<15	3,297	0.01	728	0.00	-77.92
15-24	229,943	0.88	363,327	1.24	58.01
25-34	3,129,644	11.97	3,005,033	10.24	-3.98
35-44	6,885,100	26.34	6,477,990	22.08	-5.91
45-54	7,325,544	28.03	7,947,405	27.09	8.49
55-64	5,229,903	20.01	6,808,739	23.20	30.19
≥65	3,332,038	12.75	4,738,980	16.15	42.22
Total	26,135,469	100	29,342,202	100	12.27

Source: BPS, 2013 and BPS, 2023

知乎 @最遥远的路

III. Je, Indonesia iko katika nafasi ya nusu-koloni?

Nchi ya nusu-koloni ni nchi huru kirasmi lakini yenye uhuru usio kamili. Mfano wa kawaida ni China 1840–1949. China ilipoteza haki gani? Kiuchumi, mbili kuu: uhuru wa forodha na haki ya korti ya kibalozi katika makoloni ya kigeni. Kwa upande mmoja, makampuni ya kigeni yaliweza kuagiza bidhaa nyingi China kwa ushuru wa chini sana. Kwa upande mwingine, makampuni ya kigeni katika makoloni hayakulipa ushuru kwa serikali ya China. Hizi zote zilidhoofisha ushindani wa makampuni ya China na uwezo wa kifedha wa nchi, zikizuia viwanda vya nchi ya nyuma. China ilirudisha haki za forodha na makoloni kabla na wakati wa Vita vya Pili vya Dunia, lakini machafuko ya uchumi wakati wa Vita vya Ukombozi yalifanya haki hizi zisifanye kazi.

Kwa kutumia kigezo hiki, je Indonesia ni nchi ya nusu-koloni?

Kwanza, kwa haki za forodha, Indonesia imekuwa ikirekebisha ushuru mara kwa mara^{[18][19]}.

Ili kulazimisha makampuni kutengeneza bidhaa ndani, serikali ya Indonesia hata inaziba bidhaa zao^[20]. Hata Apple haikuthubutu kuomba meli za kivita za nchi yake kulinda “biashara huru”.

为解除“封杀”，苹果妥协了！据称将在印尼投资1000万美元

原创

环球市场情报

2024-11-05 14:12 星期二 财联社 黄君芝 责编 林琦

- ①苹果公司计划在印度尼西亚投资近1000万美元，以扩大其产品生产；
- ②印尼政府此前宣布禁止在该国销售最新款iPhone 16，原因是苹果未能履行其在印尼的投资承诺。

财联社11月5日讯（编辑 黄君芝）据知情人士透露，苹果公司（Apple Inc.）计划在印度尼西亚投资近1000万美元，以扩大其产品生产，进而寻求解除印尼对其最新款iPhone的销售禁令。

这名知情人士说，该计划将涉及苹果与其供应商合作在雅加达东南部的万隆投资一家工厂。他表示，该工厂将生产苹果设备的配件和零部件等产品。印尼商务部正在审议这一提议，预计很快就会做出决定。这一提议尚未最终确定，可能会有变化。

很显然，对于印尼的“封杀”，苹果这回是妥协了。**印尼政府此前宣布禁止在该国销售最新款iPhone 16，原因是苹果未能履行其在印尼的投资承诺。**印尼政府要求苹果在印尼实现40%的国产化率，但苹果公司在印尼的工厂并未能满足这一要求。

Pili, kwa haki ya kodi, kila kampuni ya kigeni inayoingia Indonesia lazima ilipe kodi; hakuna kampuni inayojulikana kuendelea kufanya kazi bila kulipa kodi.

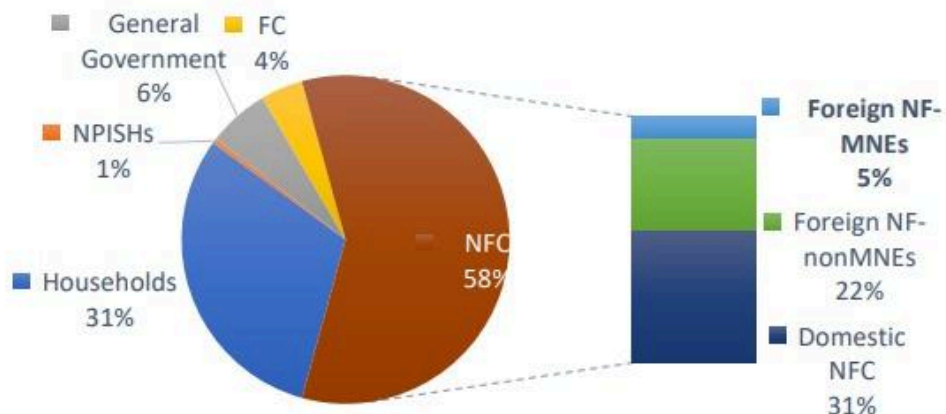
Mwisho, kwa matumizi bora ya rasilimali za taifa, serikali ya Indonesia imeanzisha mfumo wa “ubepari wa serikali”, ikilazimisha makampuni ya serikali kusafirisha rasilimali kuu^{[21][22]}.

Kwa misingi hii, tunaweza kusema Indonesia sio nchi ya nusu-koloni kabisa. Kwa maneno mengine, nchi ya nusu-koloni haiwezi kufanya mambo ambayo serikali ya Indonesia inafanya. Kwa kweli, Indonesia inaonekana kimataifa kama “nguvu ya kati”. Ikiwa nchi kama hii inaweza kuitwa nusu-koloni, basi Japan, Canada na nyingine zinaweza kuhesabiwa; dunia haitakuwa na nchi chache huru. Dhana ya nusu-koloni ingepoteza maana.

Bila shaka, wengine watasema uwiano wa mtaji wa kigeni nchini Indonesia ni mkubwa. Lakini kwa kweli, kama nchi nyingine za dunia ya tatu, Indonesia kwa muda mrefu ilikuwa na uchumi mchanganyiko na sehemu kubwa ya umma. Hata 2010, “40% ya GDP ya Indonesia ilitengenezwa na makampuni ya serikali; 2010, makampuni 141 ya serikali yalilipa kodi na faida jumla ya trilioni 120 za rupia (dola bilioni 13.5), 18% ya mapato ya serikali”^{[23][24]}. Benki za serikali bado zinashikilia zaidi ya 40% ya mali za kifedha za benki, katika nafasi ya kuongoza^[25].

2019, mchango wa makampuni ya kigeni katika GDP ya Indonesia 26%, wengine 74% kutoka makampuni ya ndani^[26]. “Jumla ya mali trilioni 59.29 za yuan, ambapo mali za makampuni ya viwanda ya serikali 41.8%, binafsi 33.1%, kigeni 25.1%”. Ikiwa faida ya mtaji wa kigeni inalingana na wa ndani, sehemu ya mtaji wa kigeni ni karibu 26%; hata ikiwa juu kidogo, inalingana na China 2010; hakuna hali ya udhibiti wa mtaji wa kigeni.

Share of foreign NF MNEs GVA to GDP 2019



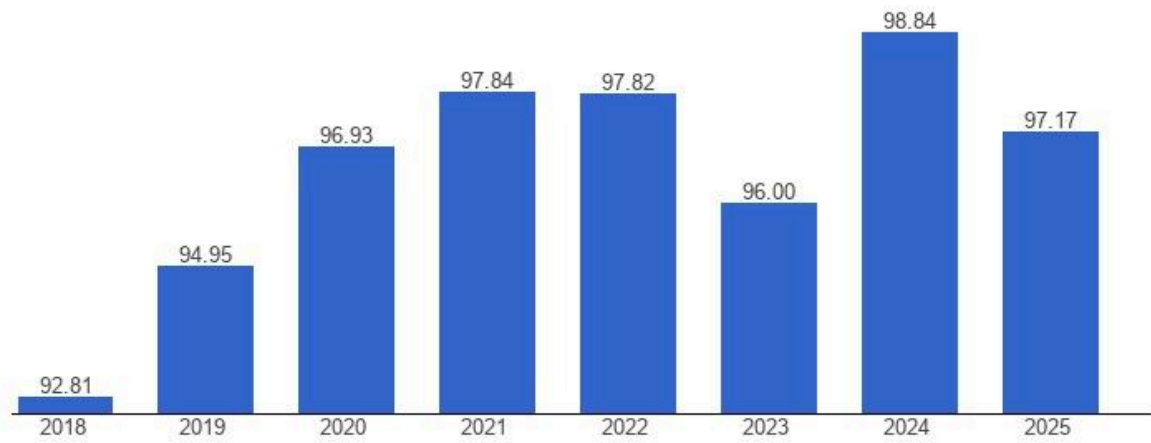
Foreign-controlled NFC contributed 26% to GDP of which foreign NF MNEs share is 5%

Kwa kweli, katika zama za utandawazi, kuingia kwa mtaji wa kigeni kunamaanisha utulivu wa kijamii na matarajio mazuri ya uchumi. Kinyume chake, nchi isiyo na mtaji wa kigeni inawezekana imetengwa na soko la dunia. Ikiwa Indonesia ni nusu-koloni, China ya miaka ya 2000 pia inaweza kuhesabiwa. Wazee wanaweza kukumbuka kwamba wakati huo baadhi ya "mabwana" wa jukwaa la Wuyou walikuwa wakihubiri nadharia hii, wakiogopa nchi yao kuwa koloni la kiuchumi la Amerika na Japan. Miaka kumi na zaidi baadaye, wakati vita vya biashara vilipoanza na mtaji wa kigeni kuanza kuondoka, labda watasema Amerika inafanya uzuiaji wa kimkakati.

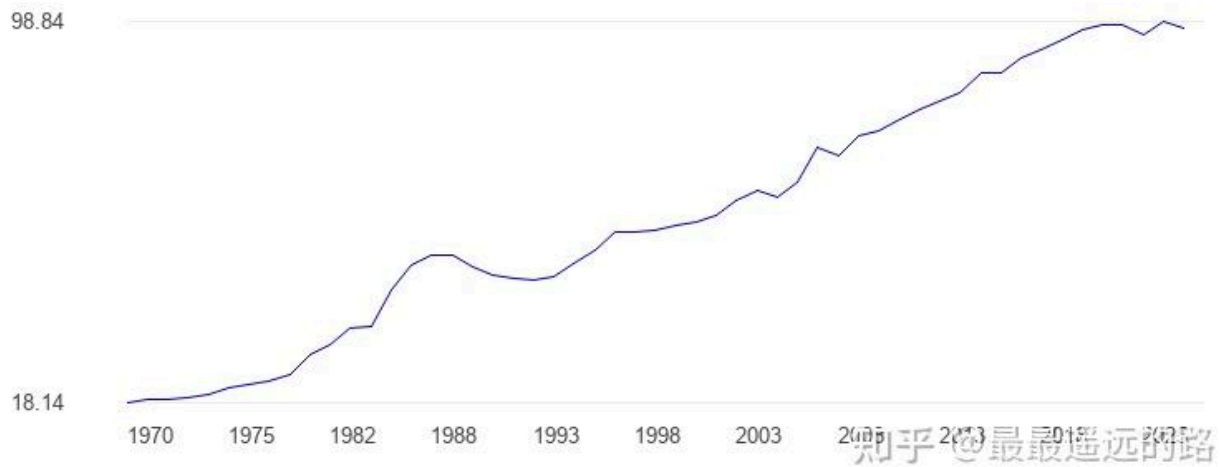
IV. Matarajio ya maendeleo

1. Nguvu kazi

Kwa sasa, uwiano wa kujiunga na shule za sekondari nchini Indonesia umefikia 97%^[27].



Historical series



Uwiano wa kujiunga na elimu ya juu wa Indonesia 2013 ulifikia 45% — juu kuliko India, hata juu kuliko Malaysia; inalingana na kiwango cha China 2014, tofauti chini ya miaka 10^[28].

School enrollment, tertiary (% gross) - Indonesia, China, India, Malaysia

Data API, UN Educational, Scientific and Cultural Organization (UNESCO), uri: databrowser.uis.unesco.org/resources, note: The data are obtained through the UIS API. Detailed documentation is available at: api.uis.unesco.org/api/public/documentation, publisher: UNESCO Institute for Statistics (UIS), type: Bulk file (csv), date accessed: 2026-03-19, date published: 2026-02

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School enrollment, tertiary (% gross) - Indonesia, China, India, Malaysia

2020, Indonesia ilipata wahitimu 300,000 wa sayansi na uhandisi — nafasi ya tano duniani^[29].

Top Countries by Number of STEM Graduates

Graduates in Science, Technology, Engineering, and Mathematics in 2020



Figure 1: Top Countries by Number of STEM Graduates in 2020. (Original graph created by Niall McCarthy at Statista)

2. Matumizi ya utafiti na maendeleo

Kwa muda mrefu, matumizi ya R&D ya Indonesia yalikuwa ya chini kabisa kati ya uchumi kuu, chini hata kuliko nchi kama India^[30].

R&D Spending in G20 nations as % of GDP		
Rank	Country	R&D spending
1	South Korea	4.93%
2	United States	3.46%
3	Japan	3.30%
4	Germany	3.14%
5	United Kingdom	2.91%
6	China	2.43%
7	France	2.22%
8	Australia	1.83%
9	Canada	1.55%
10	Italy	1.45%
11	Turkey	1.40%
12	Brazil	1.15%
13	Russia	0.94%
14	India	0.65%
15	South Africa	0.60%
16	Argentina	0.52%
17	Saudi Arabia	0.46%
18	Indonesia	0.28%
19	Mexico	0.27%
G20 Average		3.03%

Sasa, serikali ya Indonesia inapanga kuongeza matumizi ya R&D hadi 1% ya GDP ndani ya miaka michache^[31]. Mbali na fedha za umma, serikali inahimiza makampuni kuongeza R&D. Kwa mfano, Indonesia imeanzisha sera ya kupunguza kodi hadi 300% ya ziada kabla ya kodi, kuhimiza uwekezaji wa R&D katika semiconductor^[32]. Sera za viwanda, kwa kiwango chochote, zina athari fulani.

3. Maendeleo ya viwanda vya kati na vya juu

(1) Viwanda vya betri za lithiamu

Madini ya nikeli ni rasilimali muhimu kwa betri; Indonesia inashikilia 42% ya hifadhi za dunia, katika nafasi ya kuongoza kabisa. Kwa hiyo serikali ya Indonesia, kwa kutumia nafasi yake ya msambazaji mkuu wa madini, inavutia makampuni ya kimataifa kuzalisha ndani.

Kwa sasa, mradi wa mnyororo wa betri wa Indonesia (mradi wa kimkakati wa taifa PSN) unafanywa pamoja na CATL ya China^[33].

Mbali na CATL, serikali ya Indonesia pamoja na EVE Energy na Huayou Cobalt vinaandaa mradi mwingine wa betri kamili. Jumla ya dola bilioni 6^{[34][35]}.

Miradi hii miwili inajumuisha uzalishaji wa seli za betri. Kwa kulinganisha, uwezo wa seli za betri za India ni gigawati 2 tu; kwa sababu za kisiasa-kijiografia, juhudi za kuagiza mistari ya uzalishaji kutoka China zimezuwa. Kwa hiyo miradi hii ni muhimu sana kwa Indonesia: baada ya kuanza, uzalishaji wa kila mwaka utafikia dola bilioni kadhaa. Ikiwa itafanya kazi vizuri, serikali itaongeza uwekezaji na kujenga viwanda vingi vya betri.

(2) Semiconductor na nishati ya jua

Katika mnyororo wa semiconductor, Indonesia ina kiwanda cha Infineon cha kuweka na kupima kwenye kisiwa cha Batam, wafanyakazi 2,000. Lakini katika hatua nyingine, Indonesia haina uwezo wowote. Kwa hiyo Wizara ya Viwanda imeanzisha mpango wa semiconductor, kujenga uwezo wa kubuni chipu na hata uzalishaji wa wafer. Serikali inapanga kuwekeza hadi dola bilioni 1^[36]. Hazina ya ustawi wa serikali ilisaini na ARM kutoa mafunzo kwa wahandisi 15,000 wa kubuni chipu^[37].

Katika nishati ya jua, hazina ya ustawi inaongoza kikundi cha kimataifa kujenga kiwanda cha moduli za jua cha gigawati 50. Mradi umepata dola bilioni 1.4 za uwekezaji wa moja kwa moja wa kigeni^[38].

Kama nikeli, Indonesia inamiliki mchanga wa quartz, madini muhimu ya semiconductor. Hifadhi za quartz za Indonesia tani milioni 340 — zaidi ya 15% ya dunia. Serikali inapanga kukataza kabisa usafirishaji wa mchanga wa silika, kuhimiza makampuni ya jua na semiconductor kuzalisha ndani^[39].

Msingi wa Indonesia katika semiconductor na jua ni dhaifu; bila taasisi kama hazina ya ustawi, maendeleo yangekuwa magumu.

(3) Ujenzi wa meli

Indonesia ni nchi ya visiwa elfu; soko la meli ni kubwa, nguvu kazi ni nyingi; kwa hiyo kuna faida ya asili. Hazina ya ustawi Danantara inaendeleza muungano wa makampuni 22 ya serikali ya ujenzi wa meli. Februari 2026, Danantara alitangaza kuanza ukaguzi wa muungano. Akaelekeza PT PAL Indonesia kuwa kampuni mama kwa muungano na kuunda kundi la taifa la ujenzi wa meli^{[40][41]}.

Kwa kweli, ujenzi wa meli wa Indonesia una uwezo fulani: haitoshelezi mahitaji ya ndani tu, bali pia unauza nje. Meli kubwa ya kibiashara inayojengwa ndani ni meli ya shehena kavu ya daraja la Star 50, tani 50,000^[42]. Frigate ya makombora ya tani 6,000+^[43]. Kiwanda pia kinajenga meli ya kutua (LPD) kwa jeshi la Umoja wa Falme za Kiarabu, urefu mita 163, uhamisho tani 13,000^[44].

Kwa msaada kamili wa serikali na mahitaji ya ndani yanayoongezeka, ujenzi wa meli wa Indonesia utapata maendeleo makubwa.

4. Viwanda vya silaha

Viwanda vya silaha vya Indonesia ni dhaifu kuliko vya Vietnam, lakini havina msingi tu; katika zama za nguvu nyingi, Indonesia inapata fursa nyingi za kupata teknolojia kutoka nguvu kubwa na za kati. Kwa kutumia teknolojia zilizoingizwa, na kujenga juu ya nguvu za uchumi na watu, inawezekana kutengeneza vifaa vya kisasa vinavyofaa hali ya nchi.

Indonesia ilisaini makubaliano ya uhamisho wa teknolojia na China, ambapo China iliruhusu makampuni ya ndani (kama PT Pindad na PT PAL) kujenga mistari ya kukusanyia na kutunza makombora ya C-705^[45]. Mradi wa uhandisi wa kurudi nyuma, lakini matokeo hayajulikani^[46].

Katika mazoezi ya 2026, jeshi la majini la Indonesia lilitumia C-705 kushambulia malengo ya nchi kavu, ikiwezekana kumaanisha ufahamu zaidi wa kombora hili^[47]. Mlango wa Malacca.

Kwa jeshi la ardhini: vifaru vya kati vya "Harimau" vilivyotengenezwa pamoja na Uturuki na kukusanywa Indonesia, uwiano wa ujanibishaji karibu 50%^{[48][49]}. Indonesia pia imejenga uwezo wa ndani wa risasi kupitia maendeleo na uhamisho wa teknolojia^{[50][51][52]}.

Kwa ndege zisizo na rubani: "Elang Hitam" ya Indonesia, inasemekana yote ya ndani isipokuwa injini^[53], eneo la mapigano kilomita 250, muda wa angani saa 24. Indonesia pia imetengeneza ndege ya "Kamikaze Drone"^[54], mnyororo wa usambazaji isipokuwa risasi ni wa ndani. Masafa ya juu ya kushambulia kilomita 80.

Mbali na miradi ya ndani, Indonesia inaagiza makombora na ndege nyingi zisizo na rubani kutoka Uturuki kwa uzalishaji wa ndani.

Katika zama za makombora na ndege zisizo na rubani, lengo la muda mfupi la Indonesia ni kujenga uzuiaji kwenye Mlango wa Malacca na njia za Bahari ya China Kusini; lengo la muda mrefu ni kutengeneza ndege na makombora ya kilomita 2000+. Mara tu uwezo huo usio sawa ukikamilika, unaweza kutishia maeneo makuu ya viwanda ya nguvu kubwa ya kaskazini na kupunguza hatari ya uvamizi. Ndiyo, China tayari ni nguvu kubwa ya dunia; Indonesia haiwezi tena kufanya kampeni za anti-Chinese; inabidi tu kufikiria kutodhulumiwa.



V. Hitimisho

Kwa msingi wa uchambuzi wa kiwango cha viwanda, kiwango cha usafirishaji, muundo wa tabaka na nyingine, tunaweza kusema Indonesia tayari ni nchi ya kibepari karibu kukamilisha viwanda; haipaswi kuonekana kama nchi ya nusu-feudal. Wakati huo huo, serikali ya Indonesia ina uhuru kamili wa forodha, kodi na umiliki wa madini makuu; Indonesia haipaswi kuonekana kama nusu-koloni. Indonesia ni nchi ya viwanda ya “nusu-ki pembe” ya mfumo wa kibepari wa dunia, inayopokea uhamishaji mkubwa wa viwanda kutoka nchi za kaskazini na China. Katika miaka 15 ijayo, Indonesia itakamilisha viwanda na kuanza viwanda vya kati na vya juu. Kwa kuunganisha rasilimali na kutekeleza sera za viwanda kwa ufanisi, Indonesia inaendelea kwenye “njia ya ubepari wa serikali”^[55], kama Morocco na Saudi Arabia, kwa kutumia rasilimali zake kutengeneza viwanda vya ndani vyenye ushindani duniani na kiwango fulani cha viwanda vya ulinzi.

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本文原发表于知乎专栏/回答 (中文): <https://zhuanlan.zhihu.com/p/2069185424879308950>; 作者: 最最遥远的路: <https://www.zhihu.com/people/zui-zui-yao-yuan-de-lu-32/posts>